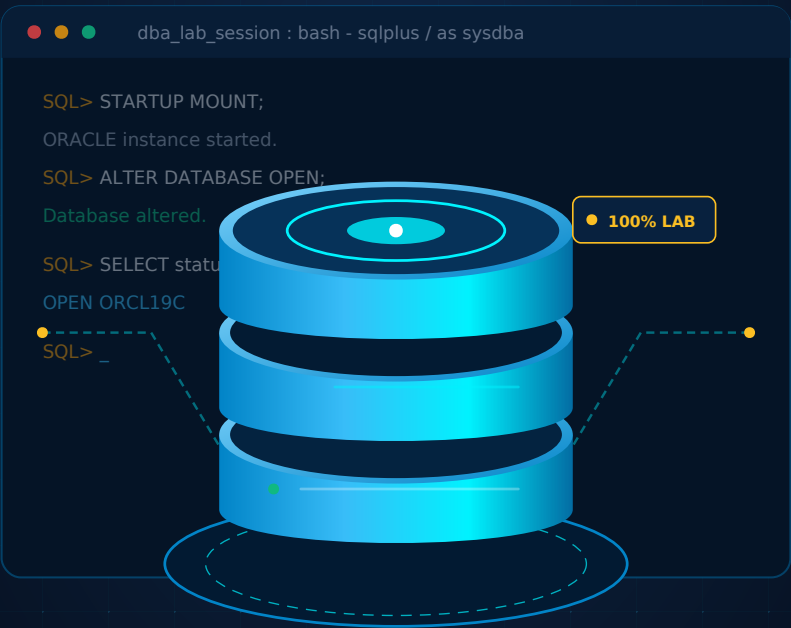


ORACLE DBA PRACTICE

An End-to-End Beginner's Guide

Practical Learning Notes from Oracle DBA Lab Practice



AUTHOR

MD SHAHID HUSSAIN

Presented By



Oracle DBA Practice

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ओरेकल डीबीए अभ्यास
शुरुआत से अंत तक शुरुआती विद्यार्थियों के लिए मार्गदर्शिका
ओरेकल डीबीए लैब अभ्यास से तैयार व्यावहारिक अध्ययन नोट्स



Md Shahid Hussain

Md Dilawar Hussain

Md Anzar Khan

Oracle DBA Practice

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Practical Learning Notes from Oracle DBA Lab Practice

Author: MD SHAHID HUSSAIN

Recommended by: Mr. Syed Shabaz

30 Years of Training Experience

Hyderabad, Telangana, India

Presented by: S³ Academy

Edition: First Edition

Publication year: 2026

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This book contains personal learning notes, laboratory commands, screenshots, and practice procedures documented during the author's Oracle DBA learning journey. The material is provided for educational and non-production practice only.

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The names, database names, usernames, hostnames, directory paths, passwords, and storage values shown in examples may be illustrative or may reflect the author's historical laboratory environment. Readers must replace them with values appropriate to their own isolated lab.

About the Author

MD Shahid Hussain holds a Master of Computer Applications (MCA) from Pune University and has over 15 years of experience across IT support, Oracle Database Administration, corporate training, trainer coordination, and business leadership. He previously served as Director & CEO of ANT Global Soft Private Limited and Amoli Jewellers Private Limited, gaining experience in entrepreneurship, organizational leadership, and client relationships.

Oracle DBA Practice grew from the notes he recorded while learning and practicing database administration in laboratory environments. It brings together commands, observations, and troubleshooting lessons to help beginners understand both the procedures and the reasoning behind them. His aim is to encourage thoughtful practice, responsible DBA habits, and the confidence to investigate problems independently.

His interests in AI Product Management and technical training extend his commitment to practical education. Through S³ Academy Training & Research Center, named after his three children—Sarim, Sarib, and Sarah—he hopes to support learners through affordable training, mentorship, and scholarships as resources and partnerships develop.

Beyond technical education, Shahid's long-term vision is to establish a school and an affordable healthcare facility for families with limited financial means. He hopes to make quality education and essential medical care more accessible, with free assistance wherever resources permit.

This book is a modest step towards that vision: a personal record of learning, shared in the hope that it will make another learner's path clearer and their next step more confident.

Author's Note

This book is based on my personal Oracle DBA learning and laboratory practice. I documented these commands, screenshots, and procedures while I was learning Oracle Database Administration and practicing in lab environments.

The purpose of this book is to share that learning journey with beginners. It is not intended to replace Oracle's official documentation, formal training, professional DBA supervision, or production change-management procedures.

Some examples in this edition use Oracle 10g, 11g, and 12c environments because those were the versions available during my original practice. A command that worked in my historical lab may behave differently in another Oracle release, operating system, edition, or configuration. Readers should therefore verify every procedure against the official documentation for the version they are using.

Please practice only in a dedicated, isolated, non-production environment. Do not copy commands directly into a live database. Before changing any system, understand the command's purpose, confirm the required privileges, take appropriate backups, and plan how the change can be reversed or recovered.

My goal is to help learners move beyond memorizing commands. A DBA must understand what a command changes, how to verify the result, how to recognize an error, and how to recover safely when the result is not as expected.

My Goals for Learners

- Build a strong foundation in Oracle Database Administration through clear explanations of essential concepts.
- Bridge theory and practice with step-by-step exercises in a dedicated lab environment.
- Encourage understanding beyond memorization by explaining each command's purpose, application, and impact.
- Develop systematic troubleshooting skills through careful observation, investigation, and learning from errors.
- Cultivate responsible DBA habits, including validation, documentation, and thoughtful execution.
- Strengthen confidence and independence for further learning, interviews, and entry-level DBA opportunities.
- Make practical knowledge accessible, especially to beginners and learners with limited financial resources.
- Inspire continuous growth through curiosity, consistent practice, and a commitment to learning.

My hope is that this book helps every reader move forward with the knowledge, confidence, and discipline to grow as an Oracle DBA.

Acknowledgements

First and foremost, I thank Almighty Allah for the strength, patience, and opportunity to bring my Oracle DBA practice notes together and share them with others.

I express my heartfelt gratitude to my parents and family for their love, sacrifices, and constant encouragement throughout my learning journey. I also extend my sincere thanks to my dear friends, Mr. Dilawar Hussain and Mr. Anzar Khan, for their friendship and unwavering support.

I am deeply thankful to my wife and children for their patience, understanding, and belief in me. They remind me of the purpose behind my efforts, and their love continues to inspire me as I work towards bringing the vision of S³ Academy to life.

My sincere thanks go to Mr. Syed Shabaz sir of Hyderabad, an Oracle DBA and Oracle Apps DBA professional trainer with 30 years of experience, for recommending this book as a resource for learning and hands-on lab practice. I deeply appreciate his encouragement, guidance, and support in helping this humble effort reach aspiring Oracle DBAs.

I am also grateful to the teachers, senior colleagues, and fellow professionals from whom I have learned throughout my career. Their advice and the practical challenges I encountered helped me develop the habit of recording commands, observing results, and learning from mistakes.

To my friends, LinkedIn connections, and readers who have shown interest in the sample, shared feedback, or helped introduce it to others: thank you. Your support has encouraged me to prepare these notes for a wider audience.

Through S³Academy, my personal learning initiative, I hope to make practical technical education more accessible, particularly for students and beginners seeking a starting point.

Please consider this book a small contribution from my learning journey to yours. If these notes help you understand a concept, practice more confidently, or ask a better question, they will have served a meaningful purpose.

With gratitude,
MD Shahid Hussain
Former Director & CEO

S³ Academy — A Personal Inspiration, A Shared Purpose

S³ Academy began with a personal inspiration: my three children—Sarim, Sarib, and Sarah. Their names share the initial “S.” Together, these three initials inspired S³, pronounced “S Cube”, and meaning S to the power of three.

The name holds a special place in my heart. It also represents a hope that reaches beyond my own family: to help students access education, encouragement, and practical skills, especially when financial difficulties limit their opportunities.

My vision is to develop S³ Academy into an educational organization that supports learners through scholarships, selected free training batches, and courses at affordable fees. These programs will be introduced gradually as resources, trainers, and partnerships become available.

Alongside technical training, we aim to provide mentorship, career guidance, and practical learning opportunities, helping students make informed choices and prepare for their next steps.

My longer-term dream is to establish a school and an affordable healthcare facility, where children from families with limited means can receive quality education and students and their families can access essential medical care at low cost. Where resources allow, I hope to extend free support to those who need it most.

This is a vision I hope to build toward, step by step, with responsible planning and the support of people who share this purpose. This book is one small beginning—a contribution from my learning journey to those starting their own.

Inspired by Sarim, Sarib, and Sarah, S³ Academy carries a simple hope: to help more families access learning, care, and the opportunity to build a better future.

MD Shahid Hussain
Founder, S³ Academy

How to Use This Book

This book is a practice companion. It is best used together with an isolated Oracle lab and the official Oracle documentation.

For each exercise:

- Read the objective before running any command.
- Confirm the Oracle version, operating system, database name, and lab topology.
- Check the required privileges and prerequisites.
- Replace example paths, usernames, database names, and passwords with safe lab values.
- Run one step at a time and record the output.
- Verify the result using an appropriate database view, command, log, or test.
- Investigate errors instead of ignoring them.
- Complete the cleanup or rollback steps.
- Record what you learned and what differed in your environment.

A command completing successfully does not necessarily mean that the objective has been achieved. A backup must be validated. A configuration change must be checked for its actual effect. A performance issue must be investigated using evidence.

Version Scope

The original practice material documented in this book was created using Oracle 10g, Oracle 11g, and Oracle 12c environments. Version-specific output and paths are retained because they are part of the author's learning record.

The examples should be read as historical lab notes unless a section explicitly states that it has been retested on a later release. Readers using another version must check differences in installation, authentication, multitenant architecture, initialization parameters, utilities, licensing, and supported features.

The book is organized to preserve the original learning journey. Advanced topics such as RMAN, ASM, Data Guard, patching, performance, and Real Application Testing may require prior knowledge, additional software, particular database editions, or a multi-host lab.

Safety Notes for Lab Practice

- Do not use the examples directly on a production or shared database.
- Do not use real passwords in scripts, screenshots, or practice notes.
- Replace passwords such as manager, S3, or other sample values with placeholders such as <LAB_PASSWORD>.
- Treat commands that terminate processes, force database shutdown, change access control, remove files, or modify storage as advanced operations.
- Do not use xhost + or similarly broad access settings unless you understand the security impact and are working in a controlled lab.
- Does not use kill -9 as a routine troubleshooting method.
- Validate backups and recovery procedures rather than assuming that a successful command proves recoverability.
- Keep a copy of the original lab notes and maintain a separate edited publication version.

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ORACLE INSTALLATION 11G

System requirement (For Practice)

For oracle 10g:-

Physical memory (RAM)	512 MB recommended
Virtual memory	Double the amount of RAM
Temp disk space	Hard disk space 1.5 GB
Video adapter	256 colors
CPU Processor	200 MHz minimum

For oracle 11g:-

Physical memory (RAM)	1GB minimum
Virtual memory	Double the amount of RAM
Temp disk space	Hard disk space 4.76 GB
Video adapter	256 colors
CPU Processor	550 MHz minimum

Create a usergroup and osuser

```
[root@S3_ACADEMY ~] # groupadd dba
```

```
[root@S3_ACADEMY ~]# df -h
```

Filesystem	Size	Used	Avail	Use%	Mounted on
/dev/sda1	15G	6.6G	7.0G	49%	/
/dev/sda5	5.7G	141M	5.3G	3%	/tmp
/dev/sda2	70G	180M	66G	1%	/u01
Tmpfs	2.0G	0	2.0G	0%	/dev/shm

```
[root@S3_ACADEMY ~] # useradd -g dba -d /u01/home -m oracle
```

-g →to specify group name

-d →to specify user's home directory

-m →to create home directory if it is not created

```
[root@S3_ACADEMY ~]# chown oracle:dba /u01/home
```

```
[root@S3_ACADEMY ~]# chmod 775 /u01/home
```

```
[root@S3_ACADEMY ~]# xhost +
```

access control disabled, clients can connect from any host

```
[root@S3_ACADEMY ~]# su - oracle
```

```
[oracle@S3_ACADEMY ~]$ whoami
```

oracle

```
[oracle@S3_ACADEMY ~]$ pwd
```

/u01/home

```
[oracle@S3_ACADEMY ~]$
```

HOW TO SHARE S/W

step 1: click on vm

step 2: select settings

step 3 select options

step 4 : select shared folders enabled

step 5 : click on always enabled

click on add
click on next
step 6 : browse (select s/w location)
click on next
step 7 : click on finish
step 8 : click on ok

THEN

```
[oracle@S3_ACADEMY ~]$ cd /mnt
```

```
[oracle@S3_ACADEMY mnt]$ ls
```

hgfs

```
[oracle@S3_ACADEMY mnt]$ cd hgfs
```

```
[oracle@S3_ACADEMY hgfs]$ ls
```

11gr2p3

```
[oracle@S3_ACADEMY hgfs]$ cd 11gr2p3/
```

```
[oracle@S3_ACADEMY 11gr2p3]$ ls
```

doc readme.html rpm sshsetup welcome.html

install response runInstaller stage

```
[oracle@S3_ACADEMY 11gr2p3]$ ./runInstaller
```

Starting Oracle Universal Installer...

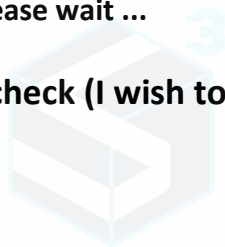
Checking Temp space: must be greater than 120 MB. Actual 5372 MB Passed

Checking swap space: must be greater than 150 MB. Actual 8001 MB Passed

Checking monitor: must be configured to display at least 256 colors. Actual 16777216 Passed

Preparing to launch Oracle Universal Installer from /tmp/OraInstall2016-09-04_11-18-55PM. Please wait ...

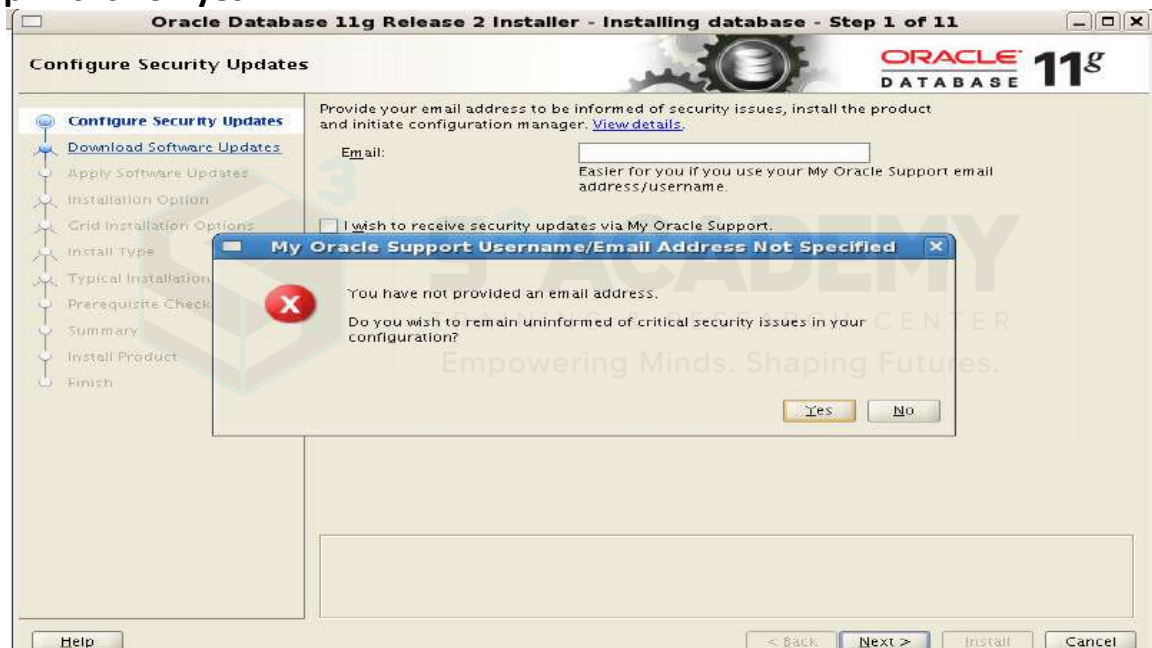
Step :1 uncheck (I wish to receive updates via my oracle support)click next



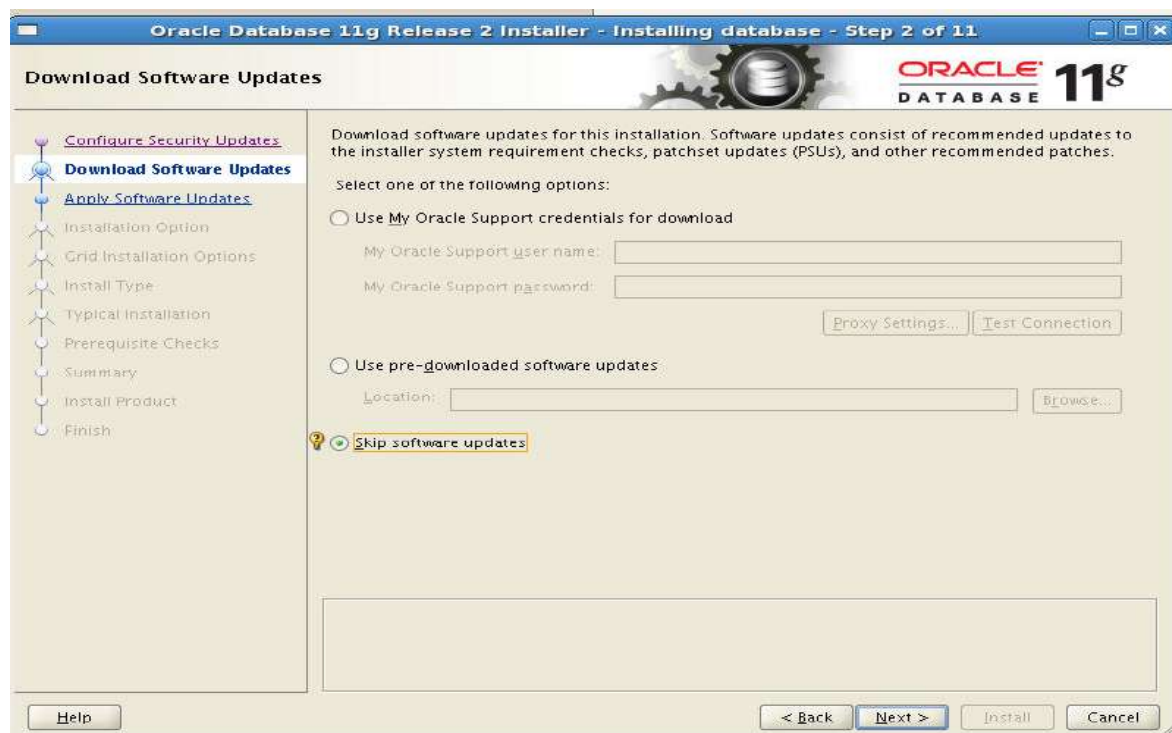
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Step :2 click on yes.



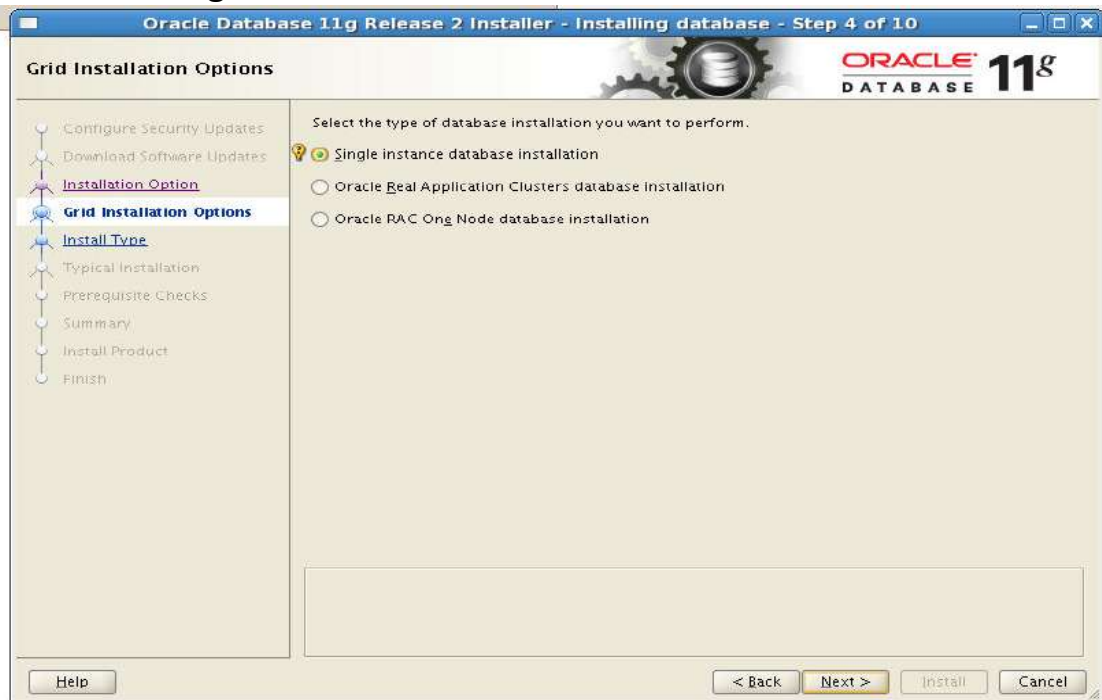
Step :3 select on skip software click next



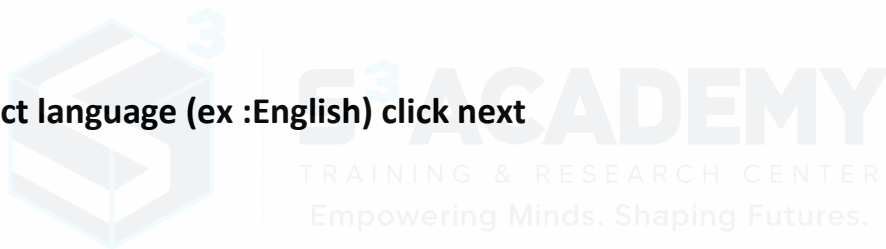
Step :4 select install database s/w only

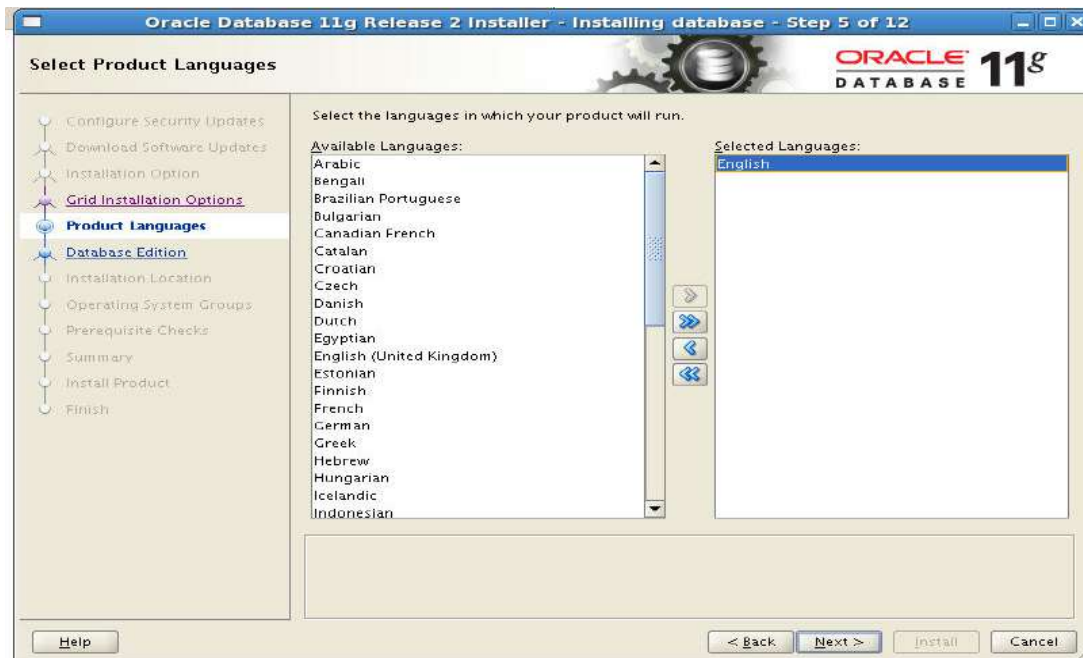


Step :5 select single instance database installation click next



Step :6 select language (ex :English) click next





Step :7 select enterprise edition click next



Step :8 give oracle base and s/w location
(ex : /u01/home/app/product/11.2.0/dbhome_1)

Click next

Specify Installation Location

Specify a path to place all Oracle software and configuration-related files installed by this installation owner. This location is the Oracle base directory for the installation owner.

Oracle Base: /u01/home/app Browse...

Specify a location for storing Oracle database software files separate from database configuration files in the Oracle base directory. This software directory is the Oracle database home directory.

Software Location: /u01/home/app/product/11.2.0/dbhome_1 Browse...

< Back Next > Install Cancel

Step :9 give inventory directory (ex: /u01/home/12c/orainventory)

Click next

Create Inventory

You are starting your first installation on this host. Specify a directory for installation files. This directory is called the "inventory directory". The installer automatically sets up subdirectories for each product to contain inventory data. The subdirectory for each product typically requires 150 kilobytes of disk space.

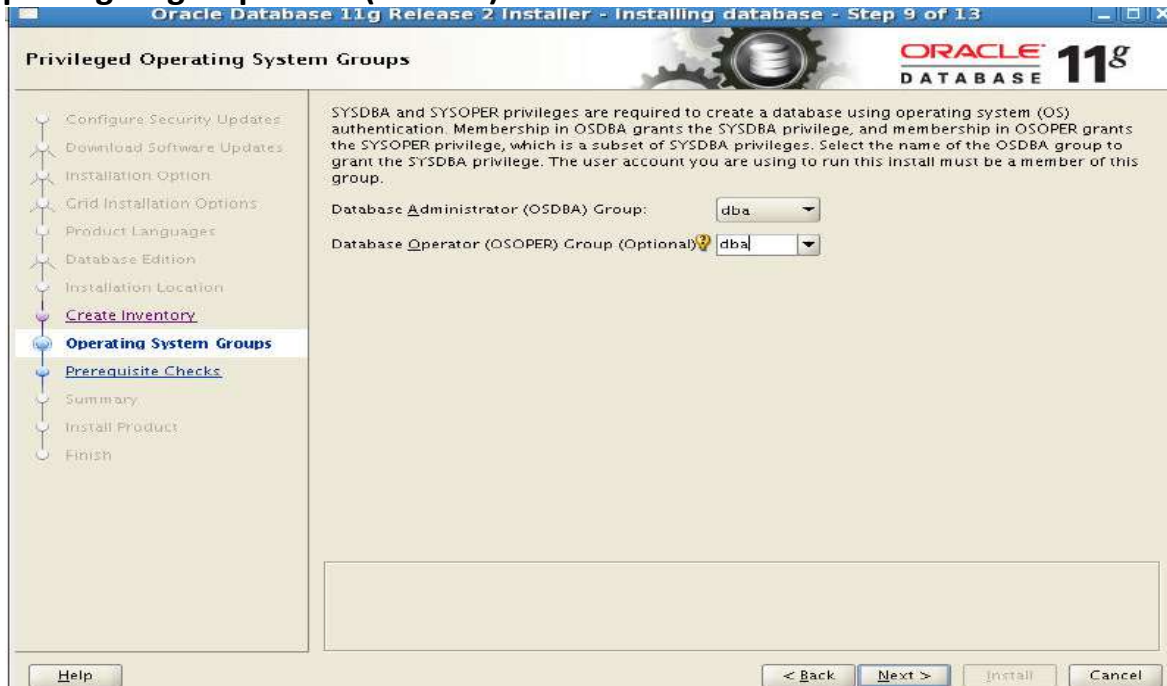
Inventory Directory: /u01/home/orainventory Browse...

Specify an operating system group whose members have write permission to the inventory directory (oraInventory).

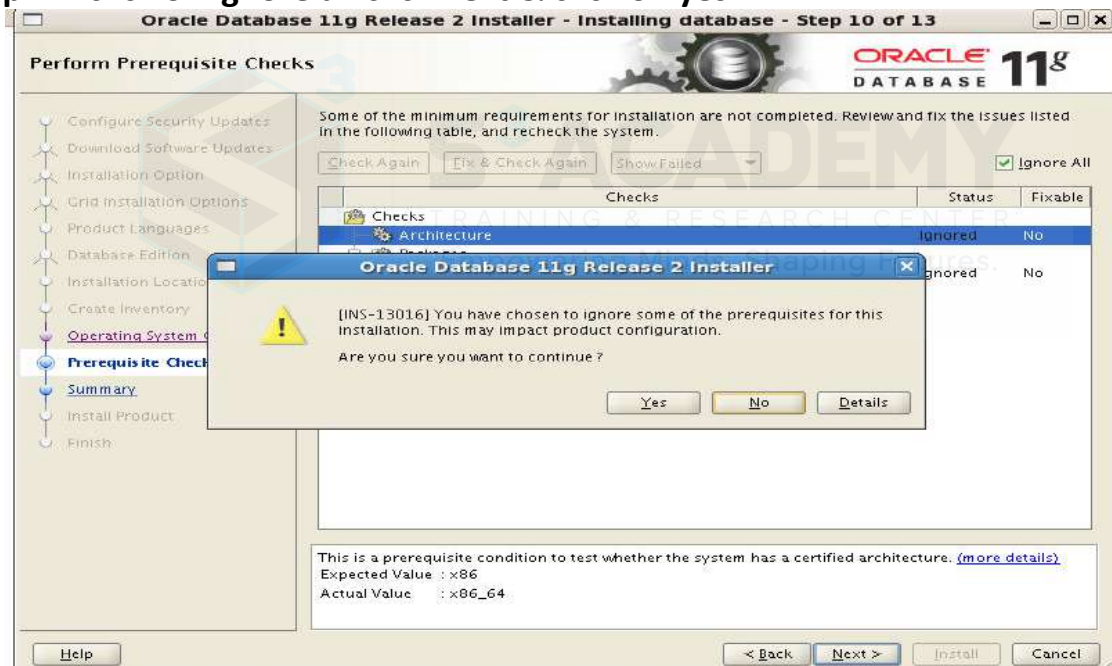
oraInventory Group Name: dba

< Back Next > Install Cancel

Step :10 give group name (ex: dba) click next



Step :11 click on ignore all click next & click on yes



Step :12 click on install



Step :13 you will get script at 94% so we have to run at root level

1:Open a new terminal

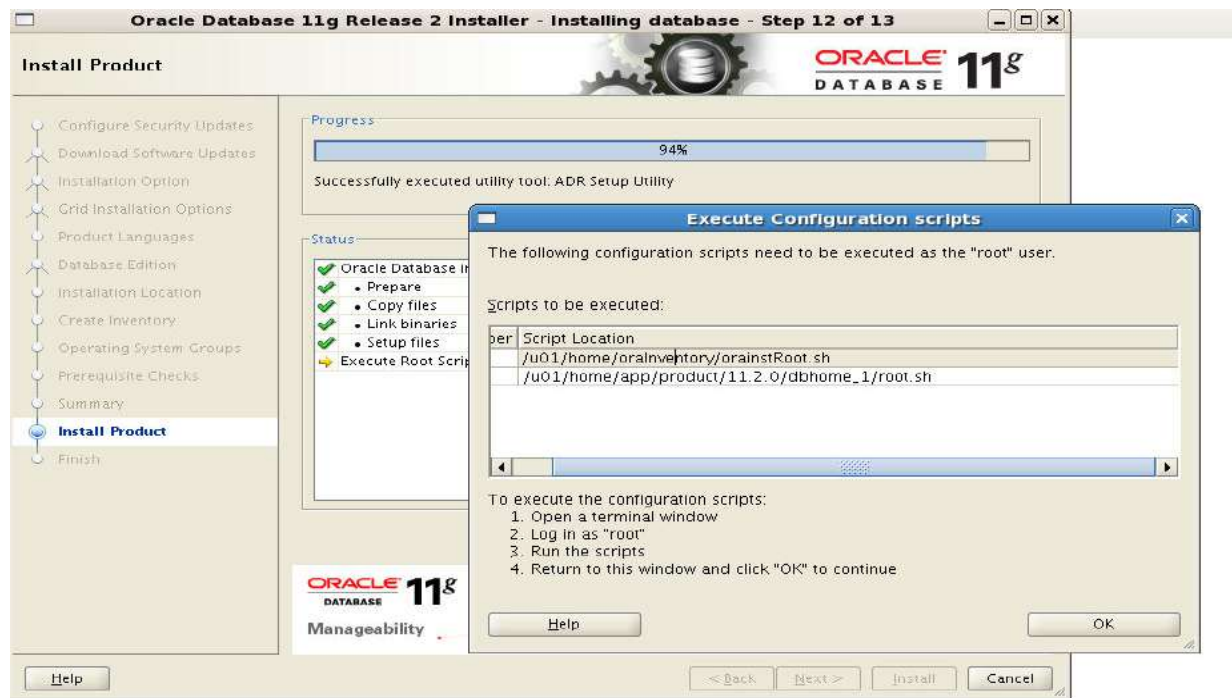
2:login as root

3:run the script

4:return to this window click ok to continue process which is at 76%

5:wait till completion of 100%

6:snapshot is mention below



[root@S3_ACADEMY ~]# /u01/home/oralInventory/orainstRoot.sh

Changing permissions of /u01/home/oralInventory.

Adding read,write permissions for group.

Removing read,write,execute permissions for world.

Changing groupname of /u01/home/oralInventory to dba.

The execution of the script is complete.

[root@S3_ACADEMY ~]# /u01/home/app/product/11.2.0/dbhome_1/root.sh

Performing root user operation for Oracle 11g

The following environment variables are set as:

ORACLE_OWNER= oracle

ORACLE_HOME= /u01/home/app/product/11.2.0/dbhome_1

Enter the full pathname of the local bin directory: [/usr/local/bin]:

Copying dbhome to /usr/local/bin ...

Copying oraenv to /usr/local/bin ...

Copying coraenv to /usr/local/bin ...

Creating /etc/oratab file...

Entries will be added to the /etc/oratab file as needed by

Database Configuration Assistant when a database is created

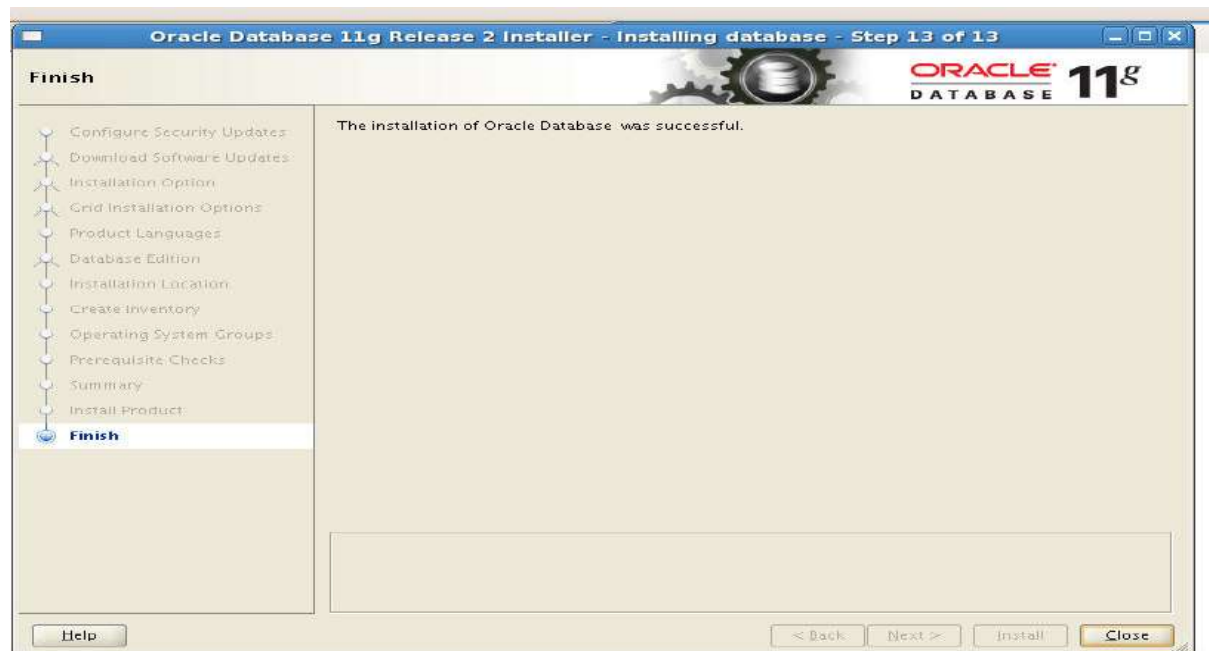
Finished running generic part of root script.

Now product-specific root actions will be performed.

Finished product-specific root actions.

[root@S3_ACADEMY ~]#

Step :15 click close .



```
[oracle@S3_ACADEMY 11gr2p3]$ cd
SET ENVIRONMENT OF ORACLE HOME (BASH PROFILE (11g))
[oracle@S3_ACADEMY ~]$ ls
app  oraInventory
[oracle@S3_ACADEMY ~]$ cd app
[oracle@S3_ACADEMY app]$ ls
checkpoints diag product
[oracle@S3_ACADEMY app]$ cd product/
[oracle@S3_ACADEMY product]$ ls
11.2.0
[oracle@S3_ACADEMY product]$ cd 11.2.0/
[oracle@S3_ACADEMY 11.2.0]$ ls
dbhome_1
[oracle@S3_ACADEMY 11.2.0]$ cd dbhome_1/
[oracle@S3_ACADEMY dbhome_1]$ pwd
/u01/home/app/product/11.2.0/dbhome_1
[oracle@S3_ACADEMY dbhome_1]$ cd
[oracle@S3_ACADEMY ~]$
[oracle@S3_ACADEMY ~]$ ls -la
. app .bash_profile .emacs .mozilla .zshrc
.. .bash_logout .bashrc .kde oraInventory
[oracle@S3_ACADEMY ~]$ vi .bash_profile
export ORACLE_HOME=/u01/home/app/product/11.2.0/dbhome_1
export PATH=$ORACLE_HOME/bin:$PATH:.
:wq!
[oracle@S3_ACADEMY ~]$ echo $ORACLE_HOME
```

```
[oracle@S3_ACADEMY ~]$  
[oracle@S3_ACADEMY ~]$ . .bash_profile  
[oracle@S3_ACADEMY ~]$ . .bash_profile  
[oracle@S3_ACADEMY ~]$ echo $ORACLE_HOME  
/u01/home/app/product/11.2.0/dbhome_1  
[oracle@S3_ACADEMY ~]$  
[oracle@S3_ACADEMY ~]$ sqlplus -v  
SQL*Plus: Release 11.2.0.3.0 Production  
[oracle@S3_ACADEMY ~]$
```



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Startup stages

Before using the oracle database we have to start the database instance.

To start the database instance we have to set instance name at os prompt.

```
$ export ORACLE_SID=<INSTANCE_NAME>
```

```
$ export ORACLE_SID=prod1
```

```
$ sqlplus / as sysdba
```

SQL*Plus: Release 11.2.0.3.0 Production on Thu Mar 24 21:15:22 2016

Copyright (c) 1982, 2011, Oracle. All rights reserved.

Connected to an idle instance.

```
SQL> startup
```

ORACLE instance started.

Total System Global Area 418484224 bytes

Fixed Size 1345352 bytes

Variable Size 138414264 bytes

Database Buffers 272629760 bytes

Redo Buffers 6094848 bytes

Database mounted.

Database opened.

When Oracle is trying to open your database, it goes through three distinct stages..

These stages are:

Nomount Stage

During the nomount stage, Oracle first opens and reads the initialization parameter file (init.ora or spfile.ora) to see how the database is configured

```
SQL> startup nomount
```

ORACLE instance started.

Total System Global Area 418484224 bytes

Fixed Size 1345352 bytes

Variable Size 138414264 bytes

Database Buffers 272629760 bytes

Redo Buffers 6094848 bytes

Mount Stage

When the startup command enters the mount stage, it opens and reads the control file.

```
SQL> startup mount
```

ORACLE instance started.

Total System Global Area 418484224 bytes

Fixed Size 1345352 bytes

Variable Size 138414264 bytes

Database Buffers 272629760 bytes

Redo Buffers 6094848 bytes

Database mounted.

If you have already started the database instance with the startup nomount command, you might change it from the nomount to mount startup stage using the alter database command:

SQL> alter database mount;

Database altered.

The Open Stage

To open the database, you can just use the startup command as seen in this example

SQL> startup

ORACLE instance started.

Total System Global Area 418484224 bytes

Fixed Size 1345352 bytes

Variable Size 138414264 bytes

Database Buffers 272629760 bytes

Redo Buffers 6094848 bytes

Database mounted.

Database opened.

If the database is mounted, you can open it with the alter database open command as seen in this example:

SQL> alter database open;

Shutting Down a Database

To shut down a database and instance, you must first connect as SYSOPER or SYSDBA.

Shutting Down with various options

SQL> shutdown normal;

SQL> shutdown; Waits for all users sessions to be disconnected

SQL> shutdown immediate; Disconnects users sessions, rollbacks uncommitted data

SQL> shutdown transactional; Waits for all users sessions transactions to be completed

- in all above options, first checkpoint is performed

SQL> shutdown abort; crashes the instance, does not perform checkpoint, no rollback

- with above abort option, instance crash recovery is required on startup